Performance Evaluation of a DNA Sequence Coding Scheme Based on Physicochemical Properties in Deep Learning

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INTRODUCTION

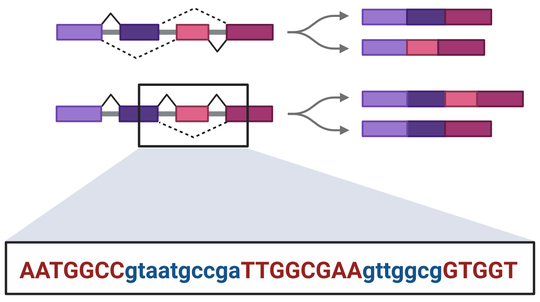
Deep Learning models, particularly Convolutional Neural Networks, are valuable for identifying patterns in genomic DNA sequences. Common encoding methods, like one-hot and K-mers, focus only on nucleotide sequences, often missing crucial structural and physical properties of DNA. By incorporating physicochemical features into sequence encoding, we aim to enhance predictive accuracy in Deep Learning models, which could significantly benefit applications in genomics and personalized medicine. Encoding DNA sequences based on physicochemical properties will lead to better model performance compared to traditional one-hot and K-mer representations.

OBJECTIVE

The main objective is to evaluate the impact of different representations, focusing on physicochemical encoding, across various convolutional network architectures for splice site prediction.

METHODS

1. Dataset

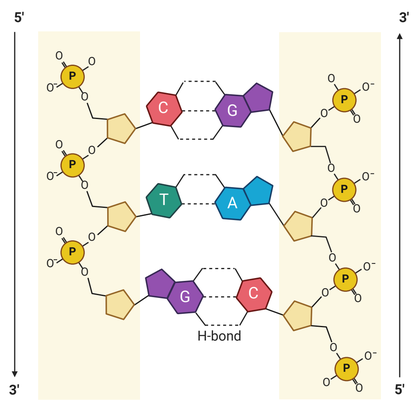


Homo Sapiens
Splice Sites - HS3D

Acceptor dataset:
6400 seqs

- 3200 True
- 3200 False

2. DNA sequences codification



Tradicional Representation:

- One-hot encoding
- k-mer vectors enconding

Physicochemical representation:

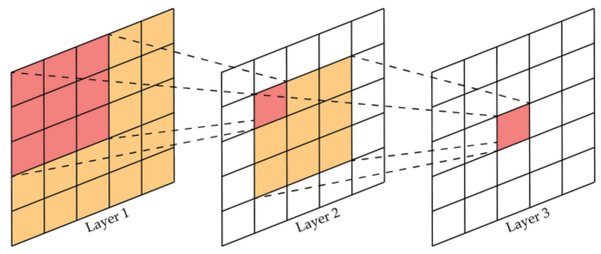
leotide	Representation
A	[(0.89), (0.94), (0.66), (1.0), (0.83), (0.90), (0.56), (0.94)]
G	[(1.0), (0.35), (1.0), (0.75), (1.0), (1.0), (1.0), (0.60)]
C	[(0.74), (0.0), (0.67), (0.5), (0.70), (0.73), (0.76), (1.0)]
T	[(0.83), (0.65), (0.67), (0.5), (0.60), (0.82), (0.87), (1.0)]

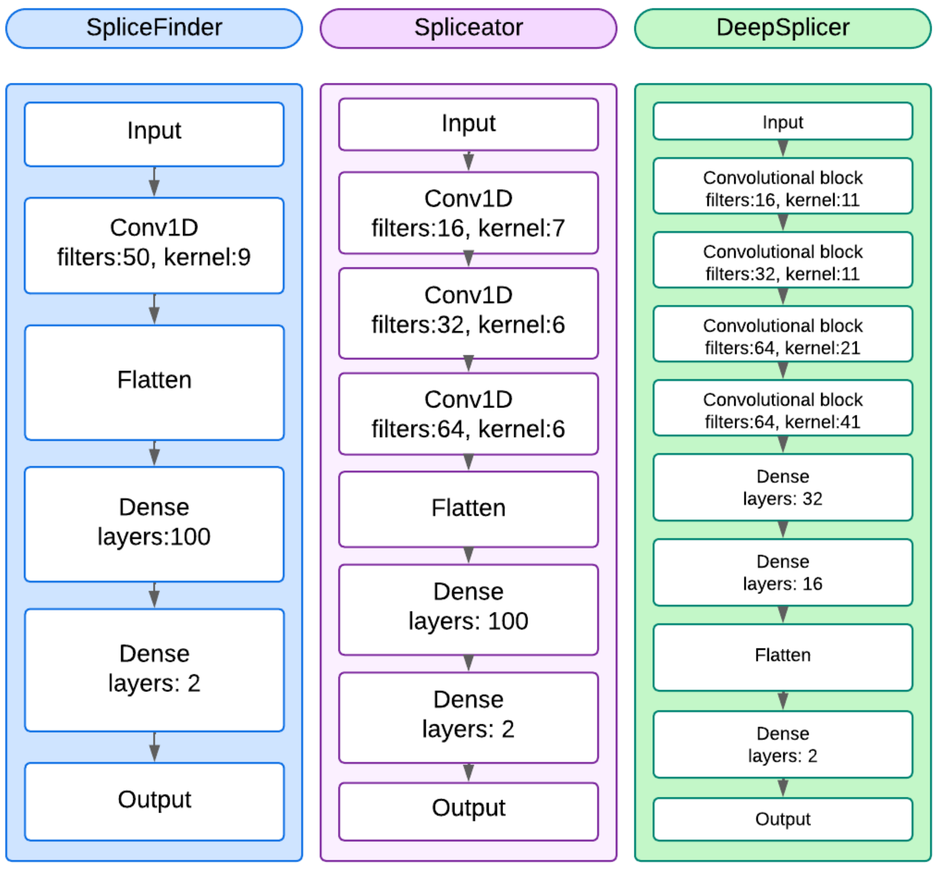
Nucleotide properties:

Molecular_Weight
XLogP3
Hydrogen_Bond
Complexity
EIIP
Heavy_Atom
Topological_Area

3. Convolutional Models

We used three model architectures to predict splice site presence, testing each architecture with three different DNA sequence representations.



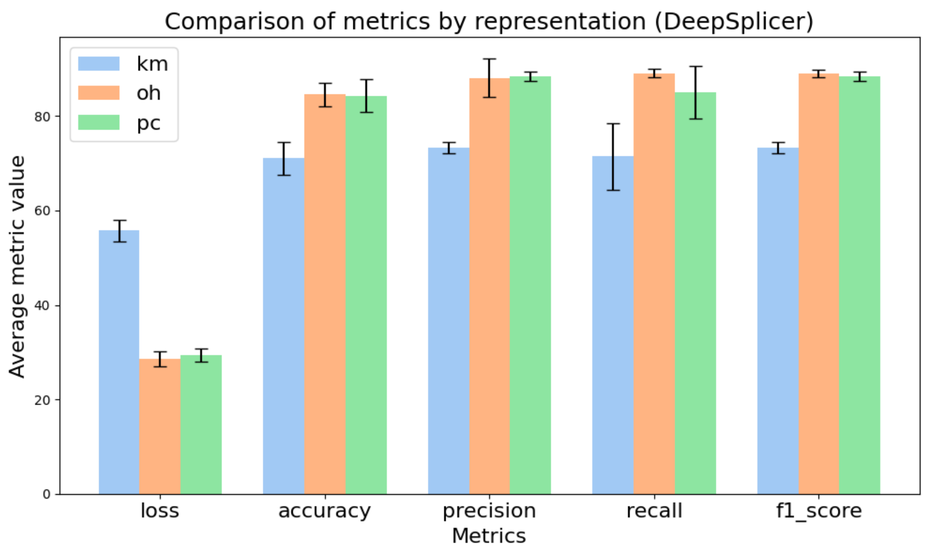


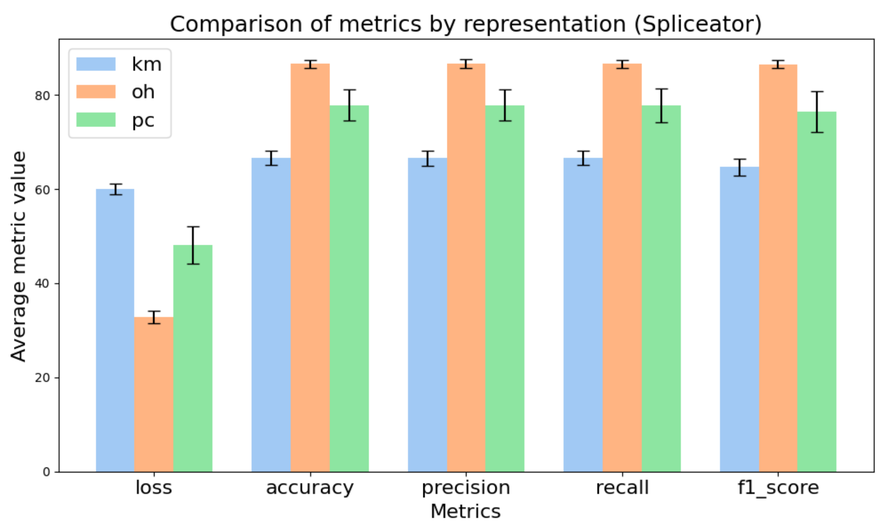
4. Performance evaluation

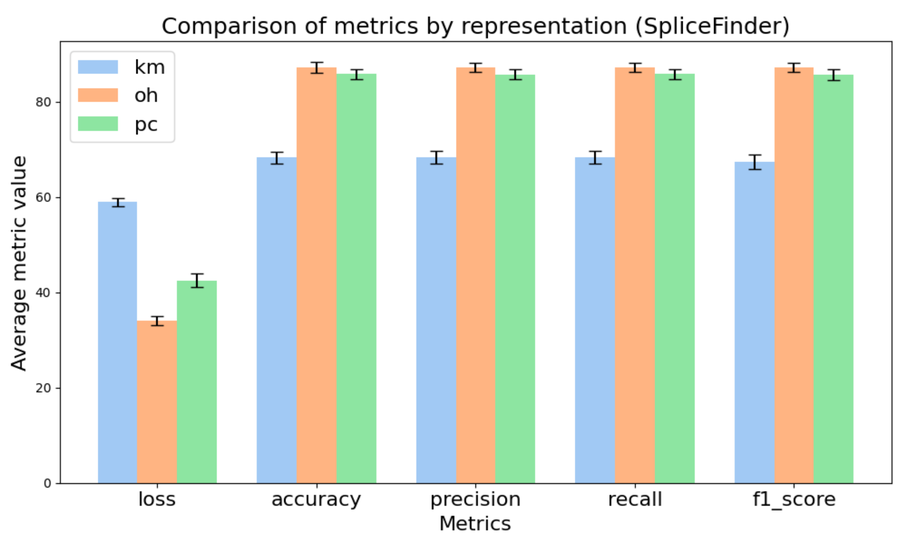
We applied 30 rounds of cross-validation with 5 epochs and a batch size of 32 to evaluate the effect of each DNA sequence representation across architectures. Confidence intervals and Mann-Whitney statistical tests were then used to compare performance metrics:

- loss
- precision
- accuracy
- recall
- f1-score

RESULTS







The one-hot encoding (oh) yielded the highest performance across all metrics, while the physicochemical representation (pc) lagged slightly behind.

With SpliceFinder Physicochemical encoding performed relatively well but did not surpass one-hot, likely due to the model's single convolutional layer, which limits complex pattern capture.

Best adaptation to physicochemical encoding was shown in Spliceator with accuracy reaching 81.09%, closer to one-hot performance than in other models. The additional layers in Spliceator enabled better feature extraction from the physicochemical representation, suggesting that a moderately deep architecture might effectively leverage complex properties in DNA sequence modeling.

CONCLUSION

Physicochemical encoding showed potential, particularly in Spliceator, where a balanced depth allowed for effective pattern recognition. However, one-hot remains the top-performing representation across architectures, especially in models with greater depth, like DeepSplicer. The results highlight that physicochemical representations may benefit from architectures with intermediate complexity or more specific, while one-hot encoding provides robust performance across all tested models.

Acknowledgment

FEnFiSDi
Fundamentos y Enseñanza de la Física y los Sistemas Dinámicos
UNIVERSIDAD DE ANTIOQUIA

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